

The background of the slide is a dark green field filled with numerous 3D cubes of varying sizes and orientations, creating a complex, isometric geometric pattern.

NEOSTATS

Winning Together with Intelligence

AI Engineering: Roles and Responsibility



Core Skill Set

- ✓ Strong foundation in Python, Machine Learning, NLP, LLM and AI application development
- ✓ Ability to build scalable AI solutions
- ✓ Strong analytical thinking, problem solving, ownership mindset, and communication skills for cross-functional collaboration



Key Responsibilities

- ✓ Design, Develop, Deploy and optimize ML models and Generative AI solutions for real-world business problems
- ✓ Build and integrate LLM-powered applications, AI agents and automation workflows
- ✓ Build Proof of Concepts (PoCs), client demos, and intelligent automation solutions

Objective

Design and build a lightweight AI-powered platform on a structured dataset. This assignment tests your ability to work across the full AI engineering stack from data analysis to a deployed application.

Your submission must include:

- ✓ Exploratory data analysis and key insights
- ✓ A machine learning model with evaluation metrics
- ✓ Conversational talk to data LLM application
- ✓ A simple user interface
- ✓ Dockerized deployment with setup instructions

Assignment Scope

01



Data Understanding & EDA

Understanding and analyzing data patterns, quality, trends, and business insights to support ML model development and decision-making.

02



NL to SQL

Natural-language questions → SQL → readable business answers via LLM-powered chatbot.

03



Machine Learning Layer

Building predictive models and intelligent decision systems to generate insights, classifications, and business outcomes from data.

04



User Interface

Multi-section UI: EDA, prediction and chatbot — end-to-end demo.

05



Dockerized Deployment

Dockerfile + docker-compose.yml — evaluator runs the full app with one command.

Functional Requirements

01 Data Understanding & EDA

- ✓ Dataset summary and feature categorization
- ✓ Data quality and missing-value analysis
- ✓ At least 5 business insights with charts

02 Machine Learning Layer

- ✓ Prediction with clean pipeline
- ✓ Handle class imbalance appropriately
- ✓ Output risk score and band

03 NL to SQL System

- ✓ Natural language → SQL → readable answers
- ✓ At least 5 working query patterns
- ✓ Prompt templates with SQL validation

Functional Requirements



Part 4: User Interface



Lightweight multi-section UI for the full platform



Sections: EDA, machine learning and chatbot



Demonstrate end-to-end workflow in one application



Part 5: Dockerized Deployment



Dockerfile — environment and dependencies



docker-compose.yml — orchestrate all services



.env.example — document required variables

Expected Deliverables



Code (GitHub Repository)

- ✓ Data processing and feature engineering pipeline
- ✓ ML training and inference with saved model artifacts
- ✓ NL to SQL module with prompts and SQL validation
- ✓ UI application code
- ✓ Dockerfile and docker-compose.yml



Documentation (README.md)

- ✓ Architecture overview or component diagram
- ✓ Step-by-step setup and run instructions
- ✓ Model selection and imbalance strategy
- ✓ Evaluation metrics and results
- ✓ Prompt engineering and token optimization
- ✓ sample outputs
- ✓ Known limitations and improvements

Evaluation Criteria

A strong submission shows depth across all areas — not just the ML model.

01	ML solution design and model quality		30%
02	LLM integration and NI to SQL capability		25%
03	Prompt Techniques, hallucination reduction and conversation memory		15%
04	EDA		15%
05	Dockerization and engineering quality		10%
06	Documentation and architecture clarity		5%

What We Look For & Submission



What We Look For



Sensible preprocessing and class imbalance handling



Reliable NL - SQL agent with at least 5 working query patterns



Clean, modular, readable code



Complete end-to-end workflow



Well-explained README over complex, hard-to-run solutions



Submission Instructions



GitHub repo link or zipped source (no venv folders)



README with setup and run instructions



.env.example with all required variables



Don't place dataset in git



Presentation on the Use case



***"We believe great things happen
when technology meets creativity."***

The dataset is provided — how you understand it, model it, and present it is entirely up to you. We are excited to see what you build.

All the best!